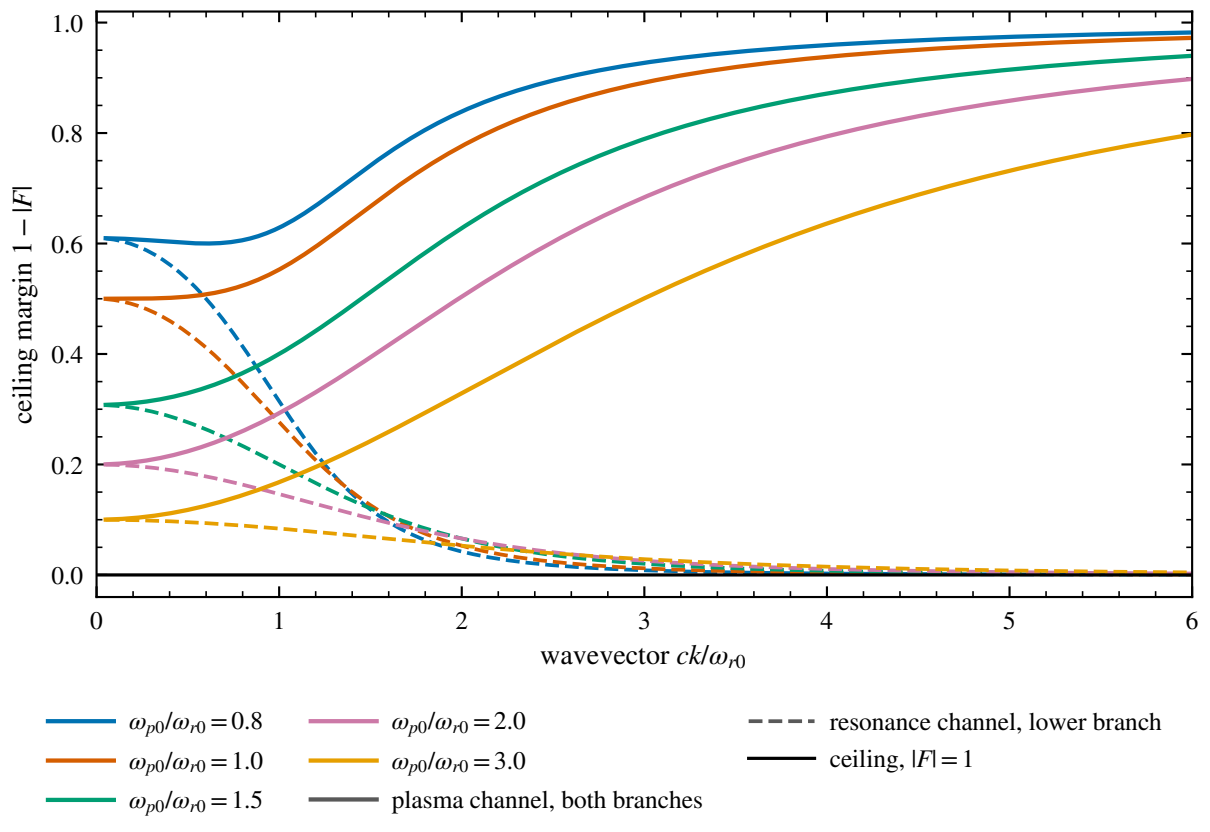
ceiling margin $1 - |F|$
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
6
wavevector ck/ω_{r0}
$\omega_{p0}/\omega_{r0} = 0.8$
$\omega_{p0}/\omega_{r0} = 1.0$
$\omega_{p0}/\omega_{r0} = 1.5$
$\omega_{p0}/\omega_{r0} = 2.0$
$\omega_{p0}/\omega_{r0} = 3.0$
plasma channel, both branches
resonance channel, lower branch
ceiling, $|F| = 1$